

Section 7.4

30. $P(t) = P_0 e^{kt}$, P = Population of bacteria, t = time

$$\therefore P(3) = 10000 = P_0 e^{3k} \quad (1)$$

$$P(5) = 40000 = P_0 e^{5k} \quad (2)$$

$$\frac{(1)}{(2)} \quad 4 = e^{2k} \rightarrow e^k = 2$$

$$\therefore 10000 = P_0 \cdot 2^3 \rightarrow P_0 = \frac{10000}{2^3} = 1250$$

34. Let $P(t) = P_0 e^{kt}$ ($P(t)$ = population at time t , P_0 = initial population, t in days)

$$\frac{dP}{dt} = P_0 k e^{kt}$$

At 8:11 PM EST, let $t = 0$ $\therefore P(0) = P_0 = 314419198$

(09/20/2012)

$$\frac{dP}{dt} = 7200 \text{ ppl/day} = P_0 k e^{kt} = P_0 k$$

$$\therefore \text{At 8:11 PM EST (09/20, 2012)} \quad \therefore k = \frac{7200}{314419198} = 2.29 \times 10^{-5} \text{ day}^{-1}$$

$$t = 365 \times 7 + 1 = 2556$$

$$P(2556) = 314419198 \times e^{(2.29 \times 10^{-5})(2556)} = 333371638$$

42. $H(t) - H_s = (H_0 - H_s) e^{-kt}$; $H_s = 18$, $H(10) = 2$, $H(20) = 10$

$$\therefore 2 - 18 = (H_0 - 18) e^{-10k} \quad (1)$$

$$10 - 18 = (H_0 - 18) e^{-20k} \quad (2)$$

$$\frac{(1)}{(2)} \quad \frac{1}{2} = e^{-10k}$$

$$\therefore H_0 - 18 = \frac{2 - 18}{e^{-10k}} = \frac{-16}{\frac{1}{2}} = -32 \quad \therefore H_0 = -14^\circ \text{C}$$

Section 9.3

13. **Salt mixture** A tank initially contains 400 L of brine in which 20 kg/L of salt are dissolved. A brine containing 0.2 kg/L of salt runs into the tank at the rate of 20 L/min. The mixture is kept uniform by stirring and flows out of the tank at the rate of 16 L/min.

- At what rate (kilograms per minute) does salt enter the tank at time t ?
- What is the volume of brine in the tank at time t ?
- At what rate (kilograms per minute) does salt leave the tank at time t ?
- Write down and solve the initial value problem describing the mixing process.
- Find the concentration of salt in the tank 25 min after the process starts.

a) Salt amount entering the tank = $0.2 \text{ kg/L} \cdot 20 \text{ L/min}$
= 4 kg/min

$$b) \frac{dV}{dt} = 20 - 16 = 4 \text{ L/min} \rightarrow V(t) = 400 + 4t$$

$$c) \frac{dy}{dt} = \text{arrival rate} - C(t) \cdot (\text{outflow rate})$$

$$= 4 - \frac{y(t)}{400 + 4t} \cdot 16$$

$$\frac{dy}{dt} + \frac{4}{100+t} \cdot y = 4$$

$$y(t) = e^{\int \frac{4}{100+t} dt} = e^{4 \ln(100+t)} = (t+100)^4$$

$$\therefore y = \frac{1}{(t+100)^4} \cdot \int 4 \cdot (t+100)^4 dt = \frac{1}{(t+100)^4} (0.8(t+100)^5 + C)$$

$$20 = \frac{1}{100^4} (0.8 \cdot 100^5) + \frac{C}{100^4} \rightarrow C = -60 \times 100^4$$

$$\therefore y = 0.8(t+100) - \frac{60 \times 100^4}{(t+100)^4} = 0.8(t+100) - \frac{60}{(1+\frac{t}{100})^4}$$

$$e) y(25) = 0.8(125) - \frac{60}{(1.25)^4} = 100 - 24.576 = 75.424$$

15. **Fertilizer mixture** A tank contains 400 L of fresh water. A solution containing 0.1 kg/L of soluble lawn fertilizer runs into the tank at the rate of 4 L/min, and the mixture is pumped out of the tank at the rate of 12 L/min. Find the maximum amount of fertilizer in the tank and the time required to reach the maximum.

$$y(0) = 0$$

$$\frac{dy}{dt} = 4 - 12 = -8 \text{ L/min}$$

$$\frac{dy}{dt} = \overset{\text{Rate in}}{0.4} - \overset{\text{Rate out}}{\frac{y \cdot 12}{400 - 8t}}$$

$$\frac{dy}{dt} + \frac{12y}{400 - 8t} = 0.4$$

$$v(t) = e^{\int \frac{12}{400 - 8t} dt} = e^{-\frac{3}{2} \ln(100 - 2t)} = (100 - 2t)^{-\frac{3}{2}}$$

$$y = (100 - 2t)^{\frac{3}{2}} \int 0.4(100 - 2t)^{-\frac{3}{2}} dt = (100 - 2t)^{\frac{3}{2}} (0.4(100 - 2t)^{-\frac{1}{2}} + C)$$

$$= 0.4(100 - 2t) - C(100 - 2t)^{\frac{3}{2}}$$

$$y(0): 0 = 0.4(100) - C(100)^{\frac{3}{2}} \rightarrow C = 0.04$$

$$\text{Max: } \frac{dy}{dt} = 0 = 0.4 - 1.2 + 0.12\sqrt{100 - 2t} = -0.8 + 0.12\sqrt{100 - 2t}$$

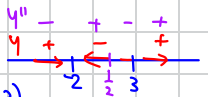
$$\sqrt{100 - 2t} = \frac{20}{3}$$

$$100 - 2t = \frac{400}{9}$$

$$t = \frac{250}{9}$$

$$\frac{d^2y}{dt^2} < 0$$

Section 9.4

1. $y = -2, 3$  $\rightarrow$ 2 Stable, 3 Unstable

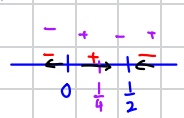
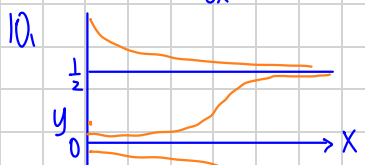
$$\frac{dy}{dx} = (y+2)(y-3)$$

$$\frac{d^2y}{dx^2} = 2y \frac{dy}{dx} - \frac{dy}{dx} = (2y-1)(y+2)(y-3)$$

7. $y = 1, 2, 3$  1, 3 Unstable, 2 Stable

$$\frac{dy}{dx} = (y-1)(y-2)(y-3)$$

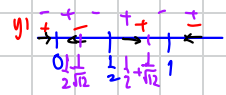
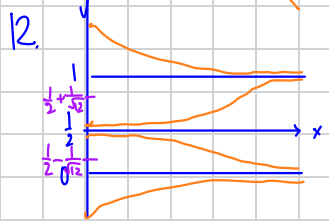
$$\frac{d^2y}{dx^2} = \frac{d}{dx}(y^3 - 6y^2 + 11y - 6) = (3y^2 - 12y + 11) \frac{dy}{dx} = (3(y-2)^2 - 1)(y-1)(y-2)(y-3)$$



$$\frac{dp}{dt} = p - 2p^2$$

$$\frac{d^2p}{dt^2} = (1 - 4p) \frac{dp}{dt}$$

$\frac{1}{2}$ Stable, 0 Unstable



$$\frac{dp}{dt} = 3p(1-p)(p - \frac{1}{2}) = 3(-p^3 + \frac{3}{2}p^2 - \frac{1}{2}p)$$

$\frac{1}{2}$: Unstable

0, 1: Stable

$$\frac{d^2p}{dt^2} = \frac{dp}{dx}(-9p^2 + 9p - \frac{3}{2}) = -\frac{3}{2}(6p^2 - 6p + 1) \cdot \frac{dp}{dt}$$

$$p = \frac{6 \pm \sqrt{36 - 24}}{12} = \frac{1}{2} \pm \frac{1}{\sqrt{2}}$$